WORKED SOLUTION

STEP III 2015, Question 1

Pure • Reduction Formulae and Substitution

STEP III 2015, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q1

The key idea

In part (i), $I_n - I_{n+1}$ combines into a single integral with u^2 on top. Writing u^2 as $u \cdot u$ and integrating by parts brings it back to I_n. In part (ii), the expression $(x - x^{-1})^2$ doesn't change when x is replaced by $1/x$, and that substitution produces the factor x^{-2}. Averaging the two forms gives $1 + x^{-2}$, which is exactly the derivative of $x - x^{-1}$, so the substitution $u = x - x^{-1}$ then works. Part (iii) is part (ii) in disguise once the denominator is written in terms of $x - x^{-1}$.

Solution

Part (i)

Combining the two integrands over a common denominator,

$$I_n - I_{n+1} = \int_0^\infty \frac{(1+u^2)-1}{(1+u^2)^{n+1}}\,\mathrm{d}u = \int_0^\infty u \cdot \frac{u}{(1+u^2)^{n+1}}\,\mathrm{d}u.$$

The second factor has an easy antiderivative:

$$\int \frac{u}{(1+u^2)^{n+1}}\,\mathrm{d}u = -\frac{1}{2n(1+u^2)^n}.$$

So we integrate by parts, differentiating u and integrating $u/(1+u^2)^{n+1}$:

$$I_n - I_{n+1} = \left[-\frac{u}{2n(1+u^2)^n}\right]_0^\infty + \frac{1}{2n}\int_0^\infty \frac{1}{(1+u^2)^n}\,\mathrm{d}u.$$

The boundary term is 0 at $u = 0$. It also tends to 0 as $u \to \infty$, because $n \geq 1$ gives $0 \leq u/(1+u^2)^n \leq u/(1+u^2) < 1/u$. Hence

$$I_n - I_{n+1} = \frac{1}{2n} I_n,$$

as required. Rearranging,

$$I_{n+1} = \frac{2n-1}{2n}\, I_n \qquad (n \geq 1).$$

For the starting value, $I_1 = [\arctan u]_0^{\infty} = \frac{\pi}{2}$. Applying the recurrence repeatedly,

$$I_{n+1} = \frac{\pi}{2} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}.$$

? **Ask yourself** How do we turn a product of odd numbers into factorials?

Multiply the top and bottom by the even product $2 \cdot 4 \cdots (2n) = 2^n n!$. The numerator then becomes $1 \cdot 2 \cdot 3 \cdots (2n) = (2n)!$, and the denominator becomes $(2^n n!)^2$. So

$$I_{n+1} = \frac{\pi}{2} \cdot \frac{(2n)!}{2^{2n}(n!)^2} = \frac{(2n)!\,\pi}{2^{2n+1}(n!)^2},$$

as required. (For a formal proof, use induction: the formula holds for $n = 0$, where it gives $I_1 = \frac{\pi}{2}$. If it holds for I_n, then multiplying by $\frac{2n-1}{2n} = \frac{(2n-1)(2n)}{(2n)^2}$ turns $\frac{(2n-2)!}{2^{2n-1}((n-1)!)^2}$ into $\frac{(2n)!}{2^{2n+1}(n!)^2}$.)

Part (ii)

First equality. Substitute $x = 1/t$ in J, so $\mathrm{d}x = -t^{-2}\,\mathrm{d}t$, and x running from 0 to ∞ corresponds to t running from ∞ to 0. Also

$$\left(x - x^{-1}\right)^2 = \left(t^{-1} - t\right)^2 = \left(t - t^{-1}\right)^2,$$

because squaring removes the sign change. Therefore

$$J = \int_{\infty}^{0} \mathrm{f}((t - t^{-1})^2)\left(-t^{-2}\right)\mathrm{d}t = \int_0^{\infty} t^{-2}\,\mathrm{f}((t - t^{-1})^2)\,\mathrm{d}t,$$

and renaming the dummy variable t as x gives the first form.

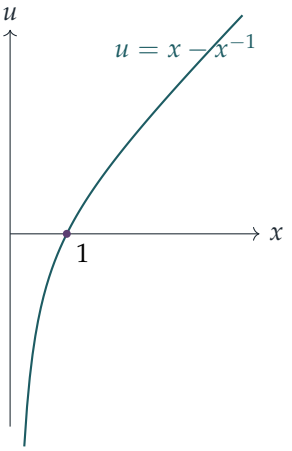
Second equality. Adding the original expression for J to this one and halving,

$$J = \frac{1}{2}\int_0^{\infty} \left(1 + x^{-2}\right)\mathrm{f}((x - x^{-1})^2)\,\mathrm{d}x.$$

Third equality. Now put $u = x - x^{-1}$, so that $\mathrm{d}u = (1 + x^{-2})\,\mathrm{d}x$. Since $\mathrm{d}u/\mathrm{d}x > 0$, u increases strictly with x, from $-\infty$ (as $x \to 0^+$) to $+\infty$ (as $x \to \infty$). So the substitution is a one-to-one change of variable on the whole range:

$$J = \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{f}(u^2)\,\mathrm{d}u = \int_0^{\infty} \mathrm{f}(u^2)\,\mathrm{d}u.$$

The last step holds because $\mathrm{f}(u^2)$ is an even function of u, so the integrals over $(-\infty, 0)$ and $(0, \infty)$ are equal. This completes part (ii).



The curve passes through $(1,0)$ and rises steadily over $x>0$, covering every real value of u exactly once.

Part (iii)

The aim is to rewrite the integrand in the form $x^{-2}\,\mathrm{f}((x-x^{-1})^2)$. Since $(x-x^{-1})^2 = x^2-2+x^{-2}$,

$$x^4-x^2+1 = x^2\left(x^2-1+x^{-2}\right) = x^2\left(1+(x-x^{-1})^2\right).$$

Hence

$$\frac{x^{2n-2}}{(x^4-x^2+1)^n} = \frac{x^{2n-2}}{x^{2n}(1+(x-x^{-1})^2)^n} = x^{-2}\,\mathrm{f}((x-x^{-1})^2), \qquad \text{where } \mathrm{f}(y) = \frac{1}{(1+y)^n}.$$

By the first and last forms of J in part (ii),

$$\int_0^\infty \frac{x^{2n-2}}{(x^4-x^2+1)^n}\,\mathrm{d}x = \int_0^\infty \frac{1}{(1+u^2)^n}\,\mathrm{d}u = I_n.$$

Replacing n by $n-1$ in the formula from part (i) (valid for $n-1\geq 0$, i.e. every positive integer n) gives

$$\int_0^\infty \frac{x^{2n-2}}{(x^4-x^2+1)^n}\,\mathrm{d}x = \frac{(2n-2)!\,\pi}{2^{2n-1}((n-1)!)^2}.$$

Checks

- For $n=1$ the answer is $\frac{\pi}{2}$, so $\int_0^\infty \frac{\mathrm{d}x}{x^4-x^2+1} = \frac{\pi}{2} \approx 1.5708$. For $n=2$ it is $\frac{\pi}{4}$ and for $n=3$ it is $\frac{3\pi}{16} \approx 0.5890$. Numerical integration of the original integrals agrees with all of these, and with the cases up to $n=8$.
- The recurrence $I_2 = \frac{1}{2}I_1 = \frac{\pi}{4}$ matches the direct calculation with $u=\tan\theta$: $I_2 = \int_0^{\pi/2}\cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{4}$.

- Part (ii) with $f(y) = e^{-y}$ says $\int_0^\infty e^{-(x-1/x)^2}\,dx = \int_0^\infty e^{-u^2}\,du = \frac{\sqrt{\pi}}{2} \approx 0.88623$. A numerical evaluation of all four forms gives this value.

Common slips

- Forgetting to check that the boundary term $u/(1+u^2)^n$ vanishes at infinity. It does here, but only because $n \geq 1$.
- Substituting $u = x - x^{-1}$ but keeping the limits 0 and ∞. The new lower limit is $-\infty$, and it is the evenness of $f(u^2)$ that brings the range back to $(0, \infty)$.
- Using the formula for I_{n+1} without shifting the index: the answer to part (iii) is I_n, not I_{n+1}.

Beyond STEP

The ratio $\frac{(2n)!}{2^{2n}(n!)^2} = \binom{2n}{n}/4^n$ is a central binomial probability, and the recurrence in part (i) is the same one that gives the Wallis product for π. The trick in part (ii) is known as the Cauchy–Schlömilch transformation. It evaluates many definite integrals that have no elementary antiderivative, including Gaussian-type integrals like the one in the checks. Siklos, *Advanced Problems in Mathematics*, has more integrals that yield to a well-chosen substitution, and Bartle and Sherbert, *Introduction to Real Analysis*, makes improper integrals and change of variables rigorous.

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